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🚀 Mastering PySpark: Your Complete Guide to 46 Essential Functions That Will Transform Your Big Data Journey! Part-1

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Mayurkumar Surani

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Collection of PySpark Functions



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🌟 Introduction: Welcome to the PySpark Universe!

Hey there, data enthusiasts! 🙌 Are you ready to embark on an exciting journey through the world of PySpark? Whether you're a seasoned data engineer or just starting your big data adventure, this comprehensive guide will be your trusty companion.

Imagine having the power to process terabytes of data with just a few lines of Python code — that's the magic of PySpark! 🌟 Apache Spark's Python API brings distributed computing to your fingertips, making it possible to handle massive datasets that would bring traditional tools to their knees.

In this deep dive, we'll explore 46 essential PySpark functions that every data professional should master. But this isn't just a boring list — we'll use real-world examples, practical scenarios, and hands-on code that you can actually use in your projects! 💡

Part -1

For part 2, please click [here](#)



Complete Function Reference Guide

Here's your roadmap to PySpark mastery, organized by category:

Data Selection & Basic Operations

1. `select()` : Column spotlight
2. `withColumn()`: Data transformer
3. `filter()/where()`: Row filters
4. `drop()`: Column eliminator
5. `alias()`: Column re-namer

Data Cleaning & Quality

6. `distinct()`: Duplicate finder
7. `dropDuplicates()`: Smart duplicate remover
8. `isNull()/isNotNull()`: Null detectors
9. `fillna()/dropna()`: Null handlers
10. `cast()`: Type converter

Aggregation & Grouping

11. `groupBy()` + `agg()`: Data summarizer
12. `collect_list()/collect_set()`: List aggregators
13. `describe()/summary()` - Quick statistics

Data Transformation

14. `orderBy()/sort()` :Data organizer
15. `limit()` : Row limiter
16. `when()` + `otherwise()`: Conditional logic
17. `explode()` : Array flattener
18. `pivot()/unpivot()` : Data reshaper

Data Integration

19. `join()` : DataFrame combiner
20. `union()/unionByName()` : Vertical combiner
21. `broadcast()` : Join optimizer

Date & Time Functions

22. `date_format()/to_date()/current_date()` : Date manipulators

String Operations

23. `substr()/concat()` : String manipulators
24. `regexp_extract()/regexp_replace()` : Regex masters

Window Functions

25. `window()` : Window creator
26. `lag()/lead()` : Row navigators
27. `ntile()` : Bucket distributor
28. `rank()/dense_rank()/row_number()` : Ranking functions

🔧 Advanced Operations

- 29. `array_contains()` : Array searcher
- 30. `explode_outer()` : Safe array exploder
- 31. `posexplode()` : Positioned exploder
- 32. `monotonically_increasing_id()` : ID generator
- 33. `input_file_name()` : File tracker

📦 Complex Data Types

- 34. `array()/struct()/map()` : Complex type creators
- 35. `map_from_arrays()` : Map builder
- 36. `to_json()/from_json()` : JSON converters
- 37. `size()` : Array length calculator

🔒 Security & Hashing

- 38. `sha2()/md5()` : Hash generators

📊 Statistical Functions

- 39. `corr()/covar_pop()/covar_samp()` : Statistical analyzers

⚡ Performance & Optimization

- 40. `repartition()/coalesce()` : Partition controllers
- 41. `cache()/persist()` : Memory optimizers
- 42. `bucketBy() + sortBy()` : Storage optimizers

🌊 Streaming & Advanced Processing

- 43. `foreach()/foreachPartition()` : Row processors
- 44. `foreachBatch()` : Batch processor

🔗 SQL Integration

- 45. `createOrReplaceTempView() + spark.sql()` : SQL bridge

📁 Sample Datasets: Our Playground for Learning

Let's create some realistic sample datasets that we'll use throughout our examples. These represent common scenarios you'll encounter in real-world data engineering:

🏢 Dataset 1: Employee Management System

```
from pyspark.sql import SparkSession
from pyspark.sql.types import *
from pyspark.sql.functions import *

# Initialize Spark
spark = SparkSession.builder.appName("PySpark_Mastery").getOrCreate()

# Employee Dataset
employee_data = [
    (1, "John", "Doe", 28, "Engineering", 75000, "2020-01-15", ["Python", "Spark", "SQL"], "john.doe@company.com", "New York"),
    (2, "Jane", "Smith", 32, "Marketing", 65000, "2019-03-20", ["Excel", "PowerBI", "SQL"], "jane.smith@company.com", "Los Angeles"),
    (3, "Mike", "Johnson", 45, "Engineering", 95000, "2018-07-10", ["Java", "Scala", "Spark"], "mike.johnson@company.com", "Chicago"),
    (4, "Sarah", "Wilson", 29, "Sales", 55000, "2021-02-28", ["Salesforce", "Excel"], "sarah.wilson@company.com", "Chicago"),
    (5, "David", "Brown", 35, "Engineering", 85000, "2019-11-05", ["Python", "AWS", "Docker"], "david.brown@company.com", "New York"),
    (6, "Lisa", "Davis", 27, "Marketing", 60000, "2020-09-12", ["Google Ads", "Analytics"], None, "Los Angeles"),
    (7, "Tom", "Miller", 41, "Sales", 70000, "2017-05-18", ["CRM", "Excel"], "tom.miller@company.com", "Chicago"),
    (8, "Anna", "Garcia", 33, "Engineering", 80000, "2020-12-01", ["React", "Node.js", "MongoDB"], "anna.garcia@company.com", "New York")
]
```

```
employee_schema = StructType([
    StructField("id", IntegerType(), True),
    StructField("first_name", StringType(), True),
    StructField("last_name", StringType(), True),
    StructField("age", IntegerType(), True),
    StructField("department", StringType(), True),
    StructField("salary", IntegerType(), True),
    StructField("hire_date", StringType(), True),
    StructField("skills", ArrayType(StringType()), True),
    StructField("email", StringType(), True),
    StructField("city", StringType(), True)
])

df_employees = spark.createDataFrame(employee_data, employee_schema)
```

🛒 Dataset 2: E-commerce Sales Data

```
# Sales Dataset
sales_data = [
    (101, 1, "2023-01-15", "Electronics", "Laptop", 1200, 1, "Online"),
    (102, 2, "2023-01-16", "Clothing", "T-Shirt", 25, 3, "Store"),
    (103, 1, "2023-01-17", "Electronics", "Mouse", 30, 2, "Online"),
    (104, 3, "2023-01-18", "Books", "Python Guide", 45, 1, "Online"),
    (105, 4, "2023-01-19", "Electronics", "Keyboard", 80, 1, "Store"),
    (106, 2, "2023-01-20", "Clothing", "Jeans", 60, 2, "Online"),
    (107, 5, "2023-01-21", "Books", "Data Science", 55, 1, "Online"),
    (108, 1, "2023-01-22", "Electronics", "Monitor", 300, 1, "Store")
]

sales_schema = StructType([
    StructField("order_id", IntegerType(), True),
    StructField("customer_id", IntegerType(), True),
    StructField("order_date", StringType(), True),
    StructField("category", StringType(), True),
    StructField("product", StringType(), True),
    StructField("price", IntegerType(), True),
    StructField("quantity", IntegerType(), True),
    StructField("channel", StringType(), True)
])

df_sales = spark.createDataFrame(sales_data, sales_schema)
```

🏢 Dataset 3: Department Information

```
# Department Dataset
dept_data = [
    ("Engineering", "Tech Tower", "Alice Johnson", 50),
    ("Marketing", "Business Center", "Bob Smith", 25),
    ("Sales", "Sales Plaza", "Carol White", 30),
    ("HR", "Admin Building", "David Lee", 15)
]

dept_schema = StructType([
    StructField("dept_name", StringType(), True),
    StructField("location", StringType(), True),
    StructField("manager", StringType(), True),
    StructField("budget", IntegerType(), True)
])

df_departments = spark.createDataFrame(dept_data, dept_schema)
```

Now let's dive into our comprehensive exploration of PySpark functions! 🚀

🔍 PART 1: Data Selection & Basic Operations

1. 🎯 select() — Your Data Spotlight

The `select()` function is like having a laser pointer that highlights exactly what you need from your data. In our employee dataset, imagine you're preparing a simple contact list:

```
# Basic column selection
df_employees.select("first_name", "last_name", "email").show()

# You can also use expressions
df_employees.select(
    col("first_name").alias("First Name"),
    col("last_name").alias("Last Name"),
    col("salary").alias("Annual Salary")
).show()
```

💡 **Real-world Application:** When building dashboards or reports, you often need only specific columns. This reduces memory usage and improves performance, especially when working with wide tables that have 100+ columns.

🔧 **Pro Tip:** Always select only the columns you need early in your data pipeline to optimize performance!

2. 🧰 `withColumn()` — The Data Transformer

This is your Swiss Army knife for data transformation! Let's create some meaningful derived columns:

```
# Add a full name column
df_employees_enhanced = df_employees.withColumn(
    "full_name",
    concat(col("first_name"), lit(" "), col("last_name"))
)

# Calculate years of experience (assuming current year is 2024)
df_employees_enhanced = df_employees_enhanced.withColumn(
    "years_experience",
    year(lit("2024-01-01")) - year(to_date(col("hire_date"), "yyyy-MM-dd"))
)

# Create salary categories
df_employees_enhanced = df_employees_enhanced.withColumn(
    "salary_category",
    when(col("salary") >= 80000, "High")
    .when(col("salary") >= 60000, "Medium")
    .otherwise("Entry Level")
)

df_employees_enhanced.select("full_name", "years_experience", "salary_category").show()
```

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    StructField("quantity", IntegerType(), True),
    StructField("channel", StringType(), True)
])
```

```
    ])  
  
    df_sales = spark.createDataFrame(sales_data, sales_schema)
```

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```
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dept_data = [  
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]  
  
dept_schema = StructType([  
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    col("last_name").alias("Last Name"),  
    col("salary").alias("Annual Salary")  
) .show()
```

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df_employees_enhanced = df_employees.withColumn(  
    "full_name",  
    concat(col("first_name"), lit(" "), col("last_name"))  
)
```

```
# Calculate years of experience (assuming current year is 2024)  
df_employees_enhanced = df_employees_enhanced.withColumn(  
    "years_experience",  
    (col("hire_date").cast("date").substr(1, 4).cast("int") - 2024).abs()  
)
```

```

    "years_experience",
    year(lit("2024-01-01")) - year(to_date(col("hire_date"), "yyyy-MM-dd"))
)
# Create salary categories
df_employees_enhanced = df_employees_enhanced.withColumn(
    "salary_category",
    when(col("salary") >= 80000, "High")
    .when(col("salary") >= 60000, "Medium")
    .otherwise("Entry Level")
)
df_employees_enhanced.select("full_name", "years_experience", "salary_category").show()

```

🔗 **Why This Matters:** Feature engineering is crucial in data science. You're creating new insights from existing data — like calculating customer lifetime value, creating age groups, or deriving seasonal patterns.

3. 🔍 **filter() and where() — Your Data Detectives**

These functions are identical twins with different names. They help you find exactly what you're looking for:

```

# Find senior engineers
senior_engineers = df_employees.filter(
    (col("department") == "Engineering") & (col("age") > 30)
)
senior_engineers.show()

```

```

# Find employees without email (data quality check)
missing_emails = df_employees.where(col("email").isNull())
missing_emails.show()
# Complex filtering with multiple conditions
high_performers = df_employees.filter(
    (col("salary") > 70000) &
    (col("department").isin(["Engineering", "Sales"])) &
    (size(col("skills")) >= 3)
)
high_performers.show()

```

🏢 **Business Impact:** These filters are essential for creating targeted marketing campaigns, identifying high-value customers, or finding data quality issues.

🔪 **PART 2: Data Cleaning & Quality Control**

4. 🗑️ **drop() — The Column Eliminator**

Sometimes less is more! Remove unnecessary columns to streamline your data:

```

# Remove sensitive information before sharing data
public_employee_data = df_employees.drop("salary", "email")
public_employee_data.show()

```

```

# Remove multiple columns at once
minimal_data = df_employees.drop("id", "hire_date", "skills")
minimal_data.show()

```

🔒 **Security Note:** This is crucial for data privacy compliance (GDPR, CCPA). Always remove PII before sharing datasets!

5. 🎯 **distinct() — The Duplicate Detective**

Find unique values across your dataset:


```
# Get unique departments
unique_departments = df_employees.select("department").distinct()
unique_departments.show()
```

```
# Get unique cities where employees work
unique_cities = df_employees.select("city").distinct().orderBy("city")
unique_cities.show()
# Count unique skills across all employees
all_skills = df_employees.select(explode(col("skills")).alias("skill")).distinct()
all_skills.show()
```

6. 🗑️ dropDuplicates() — The Smart Duplicate Remover

More sophisticated duplicate handling:

```
# Remove employees with same name and department (potential data entry errors)
clean_employees = df_employees.dropDuplicates(["first_name", "last_name", "department"])
clean_employees.show()
```

```
# Remove complete duplicates
completely_unique = df_employees.dropDuplicates()
print(f"Original count: {df_employees.count()}")
print(f"After removing duplicates: {completely_unique.count()}")
```

7. 🕵️ isNull() and isNotNull() — The Null Detectives

Data quality is paramount in any analysis:

```
# Find records with missing emails
employees_no_email = df_employees.filter(col("email").isNull())
print(f"Employees without email: {employees_no_email.count()}")
```

```
# Find complete records (no nulls in critical fields)
complete_records = df_employees.filter(
    col("email").isNotNull() &
    col("salary").isNotNull() &
    col("department").isNotNull()
)
complete_records.show()
# Data quality report
total_records = df_employees.count()
null_emails = df_employees.filter(col("email").isNull()).count()
print(f"Data Quality Report:")
print(f"Total Records: {total_records}")
print(f"Missing Emails: {null_emails} ({null_emails/total_records*100:.1f}%)")
```

8. 🛠️ fillna() and dropna() — The Null Handlers

Handle missing data intelligently:

```
# Fill missing emails with a default pattern
df_filled = df_employees.fillna({
    "email": "no-email@company.com"
})
```



```
# Fill missing values with different strategies
df_smart_fill = df_employees.fillna({
    "email": "unknown@company.com",
    "city": "Remote"
})
# Drop rows with any null values
df_complete = df_employees.dropna()
print(f"Records after dropping nulls: {df_complete.count()}")
# Drop rows only if specific columns are null
df_selective_drop = df_employees.dropna(subset=["email", "department"])
df_selective_drop.show()
```

📊 PART 3: Aggregation & Grouping — Where the Magic Happens!

9. 📊 `groupBy()` + `agg()` — The Business Intelligence Powerhouse

This combination is where PySpark truly shines for business analytics:

```
# Department-wise salary statistics
dept_stats = df_employees.groupBy("department").agg(
    avg("salary").alias("avg_salary"),
    max("salary").alias("max_salary"),
    min("salary").alias("min_salary"),
    count("*").alias("employee_count"),
    stddev("salary").alias("salary_stddev")
)
dept_stats.show()
```

```
# City-wise analysis
city_analysis = df_employees.groupBy("city").agg(
    count("*").alias("employee_count"),
    avg("age").alias("avg_age"),
    collect_list("department").alias("departments_present")
)
city_analysis.show(truncate=False)
# Advanced aggregation with multiple grouping levels
detailed_stats = df_employees.groupBy("department", "city").agg(
    count("*").alias("count"),
    avg("salary").alias("avg_salary"),
    max("age").alias("oldest_employee")
).orderBy("department", "city")
detailed_stats.show()
```

📦 **Business Value:** These aggregations power executive dashboards, help identify salary gaps, and inform strategic decisions about office locations and team compositions.

10. 📋 `collect_list()` and `collect_set()` — The List Builders

Aggregate individual values into collections:

```
# Get all skills by department
skills_by_dept = df_employees.groupBy("department").agg(
    collect_set(explode(col("skills"))).alias("unique_skills"),
    collect_list("first_name").alias("employee_names")
)
skills_by_dept.show(truncate=False)
```

```
# Create a comprehensive department profile
dept_profiles = df_employees.groupBy("department").agg(
    count("*").alias("headcount"),
    avg("salary").alias("avg_salary"),
    collect_list("city").alias("cities"),
    collect_set(explode(col("skills"))).alias("skill_set")
)
```

```
)
dept_profiles.show(truncate=False)
```

📦 PART 4: Data Transformation & Manipulation

11. 📊 orderBy() and sort() — The Data Organizers

Sorting is fundamental for analysis and presentation:

```
# Sort employees by salary (highest first)
top_earners = df_employees.orderBy(col("salary").desc())
top_earners.select("first_name", "last_name", "salary", "department").show()
```

```
# Multi-level sorting
organized_data = df_employees.orderBy(
    col("department").asc(),
    col("salary").desc()
)
organized_data.select("department", "first_name", "salary").show()
# Sort by complex expressions
df_employees.orderBy(
    size(col("skills")).desc(), # Most skilled first
    col("age").asc()           # Then by age
).select("first_name", "skills", "age").show(truncate=False)
```

12. 🎯 limit() — The Data Sampler

Control the amount of data you're working with:

```
# Get top 3 highest paid employees
top_3_earners = df_employees.orderBy(col("salary").desc()).limit(3)
top_3_earners.select("first_name", "last_name", "salary").show()
```

```
# Sample data for testing
sample_data = df_employees.limit(5)
sample_data.show()
```

⚡ **Performance Tip:** Use `limit()` during development to work with smaller datasets and speed up your iterations!

13. 🔄 when() and otherwise() — The Decision Makers

Create sophisticated conditional logic:

```
# Create employee performance categories
performance_df = df_employees.withColumn(
    "performance_tier",
    when(col("salary") >= 90000, "Excellent")
    .when(col("salary") >= 70000, "Good")
    .when(col("salary") >= 50000, "Average")
    .otherwise("Needs Improvement")
).withColumn(
    "seniority_level",
    when(col("age") >= 40, "Senior")
    .when(col("age") >= 30, "Mid-level")
    .otherwise("Junior")
)
```

```
performance_df.select("first_name", "salary", "performance_tier", "age", "seniority_level").show()
# Complex business logic
```

```
bonus_calculation = df_employees.withColumn(
    "bonus_eligible",
    when(
        (col("department") == "Sales") & (col("salary") < 60000),
        col("salary") * 0.15
    ).when(
        (col("department") == "Engineering") & (size(col("skills")) >= 3),
        col("salary") * 0.10
    ).otherwise(col("salary") * 0.05)
)
bonus_calculation.select("first_name", "department", "salary", "bonus_eligible").show()
```

14. 🌟 explode() — The Array Flattener

Transform array columns into individual rows:

```
# Explode skills to analyze individual skill distribution
skills_exploded = df_employees.select(
    "first_name",
    "last_name",
    "department",
    explode(col("skills")).alias("individual_skill")
)
skills_exploded.show()
```

```
# Analyze skill popularity
skill_popularity = skills_exploded.groupBy("individual_skill").agg(
    count("*").alias("skill_count"),
    collect_list("department").alias("departments_using")
).orderBy(col("skill_count").desc())
skill_popularity.show(truncate=False)
# Find employees with specific skills
python_experts = skills_exploded.filter(col("individual_skill") == "Python")
python_experts.show()
```

🔗 PART 5: Data Integration & Joining

15. 💖 join() — The Data Matchmaker

Joins are the backbone of relational data analysis:

```
# Inner join employees with departments
employee_dept_info = df_employees.join(
    df_departments,
    df_employees.department == df_departments.dept_name,
    "inner"
)
```

```
employee_dept_info.select(
    "first_name", "last_name", "department", "location", "manager"
).show()
# Left join to keep all employees even if department info is missing
all_employees_with_dept = df_employees.join(
    df_departments,
    df_employees.department == df_departments.dept_name,
    "left"
)
# Analyze sales data with employee information
sales_with_employee = df_sales.join(
    df_employees,
    df_sales.customer_id == df_employees.id,
    "inner"
)
sales_analysis = sales_with_employee.groupBy("first_name", "last_name").agg(
    sum("price").alias("total_spent"),
    count("*").alias("order_count"),
)
```

```
avg("price").alias("avg_order_value")
).orderBy(col("total_spent").desc())
sales_analysis.show()
```

16. 📦 union() and unionByName() — The Data Stackers

Combine DataFrames vertically:

```
# Create additional employee data
new_employees_data = [
    (9, "Alex", "Turner", 26, "Marketing", 58000, "2023-01-15", ["SEO", "Content"], "alex.turner@company.com", "Boston"),
    (10, "Emma", "Watson", 31, "Engineering", 88000, "2022-06-20", ["Python", "ML", "TensorFlow"], "emma.watson@company.com")
]
```

```
df_new_employees = spark.createDataFrame(new_employees_data, employee_schema)
# Union all employees
all_employees = df_employees.union(df_new_employees)
print(f"Total employees after union: {all_employees.count()}")
# unionByName is safer when column order might differ
all_employees_safe = df_employees.unionByName(df_new_employees)
all_employees_safe.show()
```

📅 PART 6: Date & Time Mastery

17. 📅 Date Functions — Time Travelers

Working with dates is crucial in most business applications:

```
# Convert string dates to proper date types
df_with_dates = df_employees.withColumn(
    "hire_date_formatted",
    to_date(col("hire_date"), "yyyy-MM-dd")
).withColumn(
    "current_date",
    current_date()
).withColumn(
    "days_employed",
    datediff(current_date(), to_date(col("hire_date"), "yyyy-MM-dd"))
)
```

```
# Extract date components
df_date_analysis = df_with_dates.withColumn(
    "hire_year", year(col("hire_date_formatted"))
).withColumn(
    "hire_month", month(col("hire_date_formatted"))
).withColumn(
    "hire_quarter", quarter(col("hire_date_formatted"))
)
# Analyze hiring patterns
hiring_trends = df_date_analysis.groupBy("hire_year", "hire_quarter").agg(
    count("*").alias("hires_count")
).orderBy("hire_year", "hire_quarter")
hiring_trends.show()
# Format dates for reporting
df_formatted_dates = df_with_dates.withColumn(
    "hire_date_display",
    date_format(col("hire_date_formatted"), "MMM dd, yyyy")
)
df_formatted_dates.select("first_name", "hire_date_display", "days_employed").show()
```

🔤 PART 7: String Operations & Text Processing

18. ✂️ String Functions — Text Surgeons

Master text manipulation for data cleaning and feature extraction:

```
# Extract initials and create email domains
df_string_ops = df_employees.withColumn(
    "initials",
    concat(
        substr(col("first_name"), 1, 1),
        lit("."),
        substr(col("last_name"), 1, 1)
    )
).withColumn(
    "email_domain",
    regexp_extract(col("email"), "@(.+)", 1)
).withColumn(
    "name_length",
    length(concat(col("first_name"), col("last_name")))
)
```

```
df_string_ops.select("first_name", "last_name", "initials", "email_domain", "name_length").show()
# Advanced string operations
df_advanced_strings = df_employees.withColumn(
    "formatted_name",
    concat(
        upper(substr(col("first_name"), 1, 1)),
        lower(substr(col("first_name"), 2, 100)),
        lit(" "),
        upper(substr(col("last_name"), 1, 1)),
        lower(substr(col("last_name"), 2, 100))
    )
).withColumn(
    "skills_text",
    concat_ws(", ", col("skills")) # Join array elements with comma
)
df_advanced_strings.select("formatted_name", "skills_text").show(truncate=False)
```

19. 🔍 Regular Expressions — Pattern Masters

Use regex for sophisticated text processing:

```
# Extract information using regex
df_regex = df_employees.withColumn(
    "email_username",
    regexp_extract(col("email"), "([^\@]+\@)", 1)
).withColumn(
    "has_tech_skills",
    when(
        regexp_extract(concat_ws(", ", col("skills")), "(Python|Java|SQL|Spark)", 1) != "",
        True
    ).otherwise(False)
)
```

```
# Clean and standardize data
df_cleaned = df_employees.withColumn(
    "phone_cleaned",
    regexp_replace(col("first_name"), "[^a-zA-Z]", "") # Remove non-alphabetic characters
).withColumn(
    "city_standardized",
    regexp_replace(col("city"), "\\s+", " ") # Replace multiple spaces with single space
)
df_regex.select("first_name", "email_username", "has_tech_skills").show()
```

🏠 PART 8: Window Functions — The Advanced Analytics Arsenal

Window functions are where PySpark gets really powerful for advanced analytics:

20. 🏠 Window Operations — Analytical Powerhouses

```
from pyspark.sql.window import Window

# Define window specifications
dept_window = Window.partitionBy("department").orderBy(col("salary").desc())
overall_window = Window.orderBy(col("salary").desc())
# Ranking within departments
df_ranked = df_employees.withColumn(
    "dept_salary_rank",
    row_number().over(dept_window)
).withColumn(
    "dept_salary_dense_rank",
    dense_rank().over(dept_window)
).withColumn(
    "overall_salary_rank",
    row_number().over(overall_window)
)
df_ranked.select(
    "first_name", "department", "salary",
    "dept_salary_rank", "overall_salary_rank"
).show()
# Percentile analysis
df_percentiles = df_employees.withColumn(
    "salary_quartile",
    ntile(4).over(overall_window)
).withColumn(
    "dept_salary_quartile",
    ntile(4).over(dept_window)
)
df_percentiles.select("first_name", "salary", "salary_quartile", "dept_salary_quartile").show()
```

21. ⬅️ ➡️ lag() and lead() — Time Travellers

Access previous and next rows in your analysis:

```
# Salary progression analysis (simulating career progression)
career_window = Window.partitionBy("department").orderBy("hire_date")
```

```
df_career = df_employees.withColumn(
    "previous_hire_salary",
    lag("salary", 1).over(career_window)
).withColumn(
    "next_hire_salary",
    lead("salary", 1).over(career_window)
).withColumn(
    "salary_vs_previous",
    col("salary") - lag("salary", 1).over(career_window)
)
df_career.select(
    "first_name", "department", "hire_date", "salary",
    "previous_hire_salary", "salary_vs_previous"
).show()
```

🎉 Conclusion: Your PySpark Journey Continues!

Congratulations! 🎉 You've just completed a first part of a comprehensive tour through 46 essential PySpark functions. But this is just the beginning of your big data journey!

🔑 Key Takeaways:

1. **Start Simple:** Begin with basic operations like `select()`, `filter()`, and `groupBy()`.
2. **Think in Pipelines:** Chain operations together for efficient data processing
3. **Optimize Early:** Use `cache()`, `broadcast()`, and proper partitioning from the start
4. **Handle Data Quality:** Always check for nulls, duplicates, and data types
5. **Window Functions are Powerful:** Master them for advanced analytics

6. **SQL Integration:** Use `createOrReplaceTempView()` when SQL feels more natural

🚀 Next Steps:

- Practice with your own datasets
- Explore PySpark MLlib for machine learning
- Learn about Spark Streaming for real-time processing
- Dive into performance tuning and optimization
- Join the PySpark community and contribute!

💡 Pro Tips for Success:

1. Always start with small datasets during development
2. Use `.explain()` to understand query plans
3. Monitor Spark UI for performance insights
4. Cache strategically — not everything needs caching
5. Think distributed — avoid operations that require shuffling when possible

Remember, mastering PySpark is like learning a musical instrument — it takes practice, patience, and persistence. But once you get the hang of it, you'll be orchestrating beautiful data symphonies that can handle petabytes of information! 🎵

Keep coding, keep learning, and most importantly, keep having fun with your data! The world of big data analytics is vast and exciting, and you now have the tools to explore it confidently.

Happy Sparking! ⚡ 🔄 ✨

Created with ❤️ for the PySpark community. Remember to follow best practices and always consider data privacy and security in your implementations!

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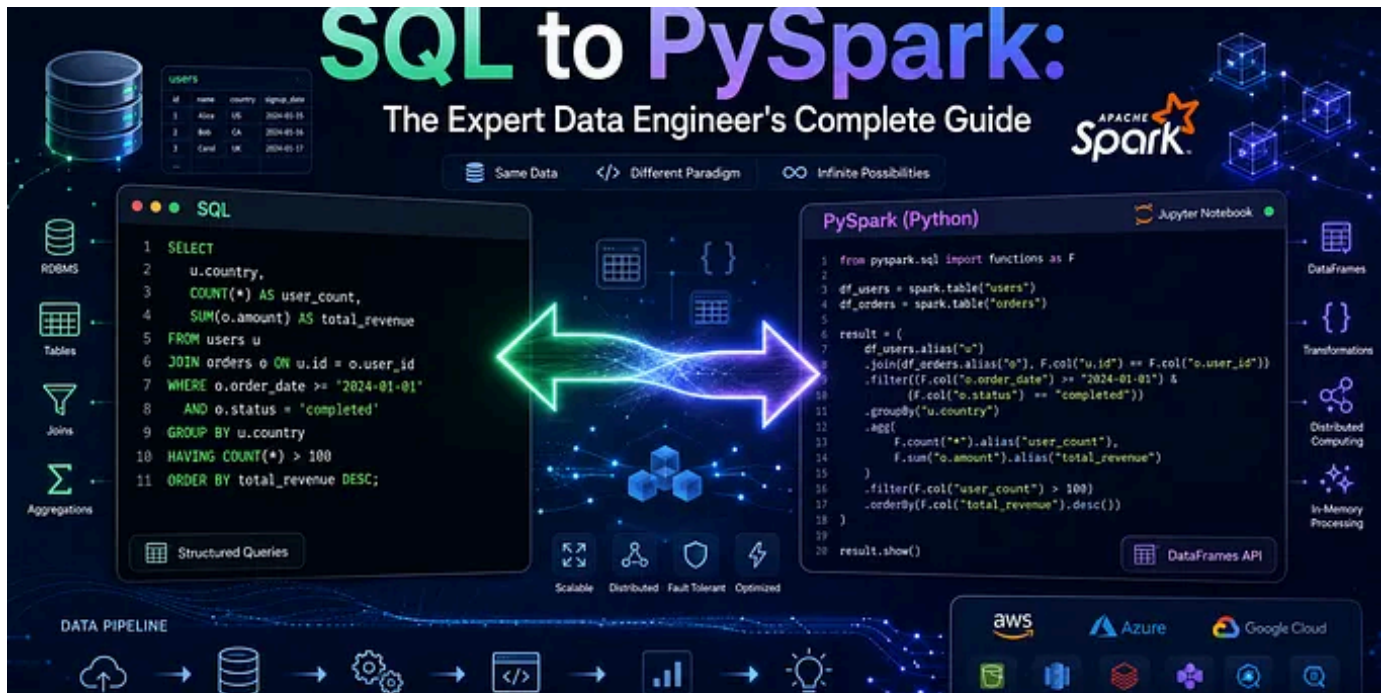
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	PySpark basics→intermediate SQL window functions, joins Code on shared screen	60–90 min	Write clean code; explain your thought process aloud
e	AWS system design Scenario-based questions Cost & security trade-offs	60–90 min	Draw diagrams; quantify your answers (TB, latency, cost)
I	Behavioural questions Project deep-dives Team collaboration	45–60 min	Use STAR method; highlight business impact of your work

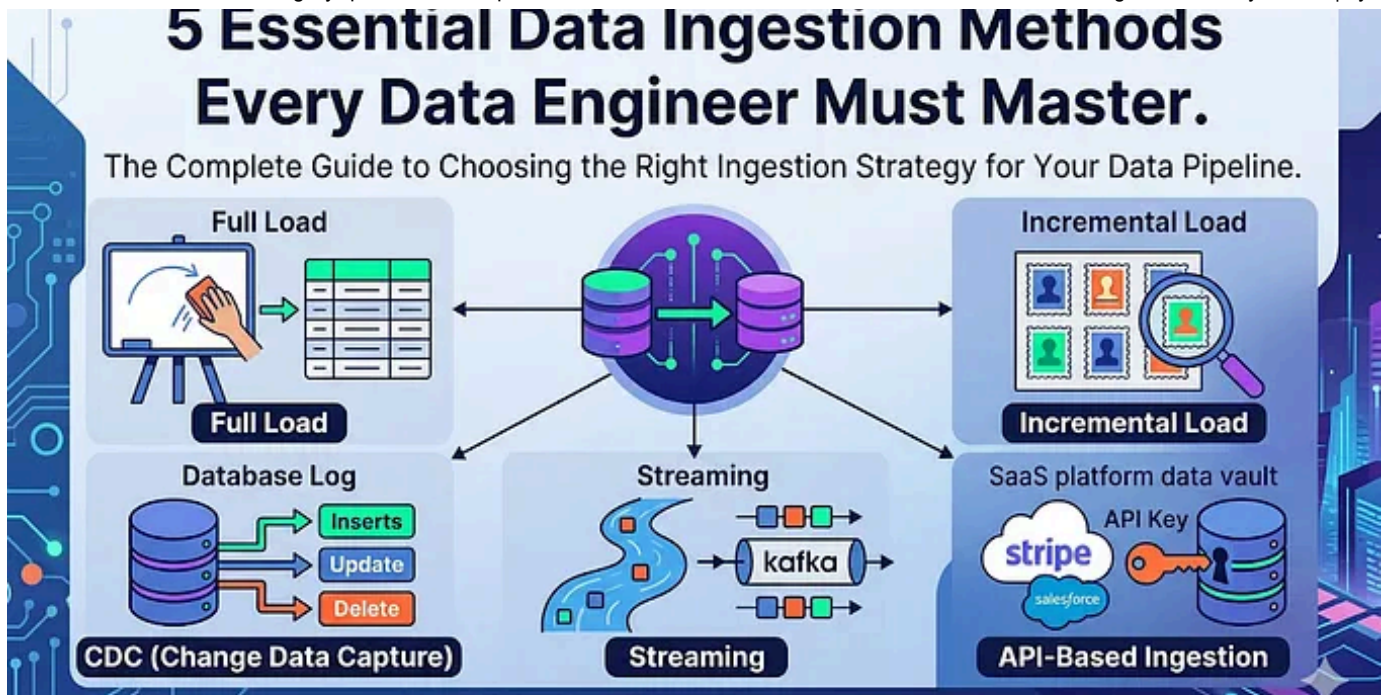
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